[*The exam text is in English. You should answer the questions in English.*]

The exam consists of 13 questions (6 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 hrs and ends at 12.00 hrs. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.
Points for each open question are indicated.

- ***Scrap paper will NOT be evaluated. Please write all your answers on the provided sheets!***
- *Please write readable and write your name and student identification number on every single sheet of paper for unique identification!*
- ***Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!***
- *Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.*
- *Good success and all the best for your studies!*

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times max\,\{\,r/\#correct\ option(s)\ -\ w/\#distractors(s), 0\,\},$$

where:
  - **5** is the total score for each question;
  - r is the number of option(s) you select correctly;
  - w is the number of option(s) you select incorrectly;
  - $\#correct\ option(s)$ is the number of correct options of this question;
  - $\#distractors(s)$ is the number of incorrect options/distractors of this question;
  - Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

No.	1	2	3	4	5	6	7	8	9	10
Answers	ACD	BC	D	AC	B	AD	AD	B	ABD	ABC

1. Regarding the biological neuron and the McCulloch-Pitts model, which of the following descriptions are correct? [more than one answer is correct]
    a. Dendrites of a biological neuron receive incoming signals, similar to the input nodes in the McCulloch-Pitts model.
    b. The axon of a biological neuron transmits signals to other neurons, which is analogous to the bias term in the McCulloch-Pitts model.
    c. Synaptic strength of the biological neuron is represented by the weights in the McCulloch-Pitts model.
    d. Spikes, or action potentials, in a biological neuron are analogous to the output of the step function in the McCulloch-Pitts model.

2. What statements are true about the stability condition for the Hopfield Network? [more than one answer is correct]
    a. A pattern is an attractor of the network if the weight on the edges are inversely proportional to the product of the node states.
    b. The condition for a stored pattern to be stable is that the node state of the evolving pattern is equal the node state of the stored pattern, for each node.
    c. Stability is obtained if $w_{ij} = \frac{1}{N}\xi_i\xi_j$.
    d. Being a stable pattern and being an attractor for a network cannot be coexisting conditions.

3. Which of the following statements is true about Hebbian learning? [one answer is correct]
    a. Hebb's rule says that the weight between two neurons should be decreased if they are activated simultaneously.
    b. The learning rate in Hebbian learning is usually larger than one.
    c. The Hebbian learning process will always stabilize after passing the training set several times.
    d. None of the above.

4. Which of the following statements are true about multi-layer perceptron (MLP)? [more than one answer is correct]
    a. MLPs consist of an input layer, one or more hidden layers, and an output layer.
    b. Each neuron in an MLP applies a linear activation function to its inputs.
    c. MLPs are capable of approximating any continuous function given sufficient neurons and layers.
    d. MLPs are inherently able to handle sequential data without modification.

5. Which of the following statements are true regarding the backpropagation algorithm? [one answer is correct]
    a. Back-propagation updates weights by minimizing the output values directly.
    b. Back-propagation requires labeled data to compute the loss function.
    c. The back-propagation algorithm operates only on feedforward neural networks.
    d. Back-propagation can be used without computing gradients.

6. Which of the following statements are true regarding the long short-term memory networks? [more than one answer is correct]
    a. Use an extended version of the back-propagation algorithm for training.
    b. Use the ReLU activation function.
    c. The functionality of the cell state is to allow vanishing gradients.
    d. LSTMs leverage gating mechanisms (input gate, forget gate, output gate) to control the flow of information and gradients.

7. Which of the following statements are true regarding RNNs? [more than one answer is correct]
    a. RNNs are capable of memorizing data.
    b. RNNs do not contain cycles in the network.
    c. The output of a RNN is a constant given a constant input.
    d. Hidden states summarize all inputs a RNN receives in the history.

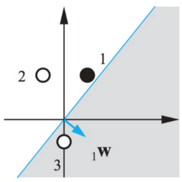
8. Which of the following tasks would be more suitable to use convolution operation? [one answer is correct]
    a. Forecast the probability of raining, provide a weather history.
    b. Classify images containing dogs from cats.
    c. Solving the XOR problem.
    d. Generate text from images (figure captioning)

9. Which of the following statements are true about the image data? [more than one answer is correct]
    a. Image data are usually represented by tensors/matrix.
    b. Colored image input also contains three (RGB) channels.
    c. Flattening the image incurs no information loss.
    d. Performing pooling operation on images incurs information loss.

10. Please calculate the size of the output image (assuming the stride is two) for a given input image and kernel size. Case (1): input size 8x8, kernel size 4x4; Case (2): input size 15x15, kernel size 5x5. [more than one answer is correct]
    a. For case 1, if no padding, the size of the output matrix is 3x3.
    b. For case 2, if no padding, the size of the output volume is 6x6.
    c. For case 1, if the padding size is (1, 1), the size of the output volume is 4x4.
    d. For case 2, if the padding size is (2, 2), the size of the output volume is 7x7.

Part II - Open Questions [50 points]

1. (**15 points**) Consider a perceptron $O = \mathrm{sgn}(\vec{w}^{\mathrm{T}}\vec{x})$ (without the bias term) and binary classification task depicted below [target label: black = 1, white = -1].



   a. (**5**) Now we wish to train this perceptron with Hebb's rule. Please write down Hebb's rule mathematically [hint: don't forget the learning rate]

The update formula is:

$$w_j' = w_j + \Delta_j$$

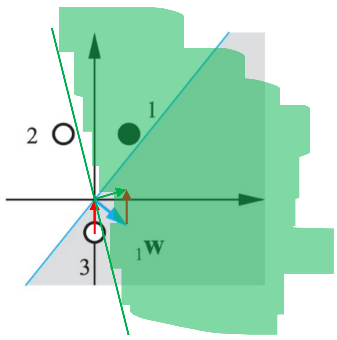
where

$$\Delta_j = \begin{cases} 2\eta y_p x_j^p & if \;\; O_p \neq y_p \\ 0 & otherwise \end{cases}$$

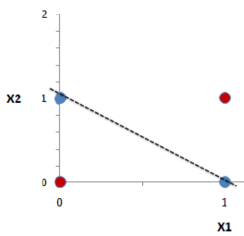
Other possible choices for Δ_j are (all equivalent – verify that):

$$\Delta_i = \eta(1 - O_n y_n)x_i^p y_n \qquad \text{and} \qquad \Delta_i = \eta(y_n - O_n)x_i^p$$

   b. (**5**) Imagine the initial weight and the decision boundary are set according to the above diagram. Note that the shaded half space is classified to the **black** label. Please demonstrate the Hebb's rule visually in this diagram by considering **point #3** [please draw a new figure below].



   c. (**5**) Now consider the famous eXclusive OR (XOR) problem. Can this problem be solved by a perceptron theoretically? Please picture this XOR problem and argue in this picture why this problem can/cannot be solved by perceptron.



XOR

X1	X2	Y
0	0	0
0	1	1
1	0	1
1	1	0

2. (**20 points**) Consider a Hopfield network, whose main task is to address the associative memory problem given multiple stored patterns (store a set of p patterns $\vec{\xi}^1, \dots, \vec{\xi}^p$ such that when presented with a new pattern $\vec{x}$, the network responds by producing whichever one of the stored patterns that most closely resembles $\vec{x}$) and perform the following tasks.

   a. (**5**) Write the generalized Hebbian rule in the case of multiple stored patterns.

How do we get the system to restore the most similar pattern to the test one out of many patterns $\vec{\xi}^1, \dots, \vec{\xi}^p$?

We make w_{ij} a SUPERPOSITION OF TERMS: $\boxed{w_{ij} = \frac{1}{N}\sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu}$ (•)

$\Rightarrow$ the network state of $\vec{x}$ will evolve to restore the closest pattern $\vec{\xi}^\mu$.

(•) is called GENERALIZED HEBBIAN RULE ("Networks which fire together, wire together")

   b. (**10**) Prove the stability condition for a pattern $\vec{\xi}^\nu$.

STABILITY CONDITION for a pattern ξ_i^ν: $\quad O_i = sgn(h_i^\nu) = \xi_i^\nu \quad \forall i = 1 \dots N$

$$h_i^\nu = \sum_{j=1}^{N} w_{ij}\xi_j^\nu = \frac{1}{N}\sum_j\sum_\mu \xi_i^\mu \xi_j^\mu \xi_j^\nu =$$

$$= \frac{1}{N}\Big(\sum_j \xi_i^\nu \underbrace{\xi_j^\nu \xi_j^\nu}_{=1} + \sum_j\sum_{\mu\neq\nu} \xi_i^\mu \xi_j^\mu \xi_j^\nu\Big) =$$

$$= \frac{1}{N}\underbrace{\sum_{j=1}^{N}\xi_i^\nu}_{=N\xi_i^\nu} + \frac{1}{N}\sum_{j=1}^{N}\sum_{\mu\neq\nu} \xi_i^\mu \xi_j^\mu \xi_j^\nu =$$

$$= \xi_i^\nu + \underbrace{\frac{1}{N}\sum_j\sum_{\mu\neq\nu} \xi_i^\mu \xi_j^\mu \xi_j^\nu}_{\equiv A}$$

now, if $|A| < 1$, then $O_i = sgn(h_i^\nu) = sgn(\xi_i^\nu + A) = sgn(\xi_i^\nu) = \xi_i^\nu$

$\Rightarrow$ STABILITY.

   c. (**5**) Please define the crosstalk term.

$$\underbrace{\frac{1}{N}\sum_j\sum_{\mu\neq\nu} \xi_i^\mu \xi_j^\mu \xi_j^\nu}_{\equiv A}$$

$\rightarrow$ can be also seen as

$$\frac{1}{N}\sum_{\mu\neq\nu}\xi_i^\mu \underbrace{\sum_j \xi_j^\mu \xi_j^\nu}_{\vec{\xi}^\mu\cdot\vec{\xi}^\nu}$$ CROSSTALK

3. (**15 points**) Consider the convolution and pooling operations and answer the following questions:
   a. (**5**) Compute the resulting feature map using a stride of 2 and no padding.

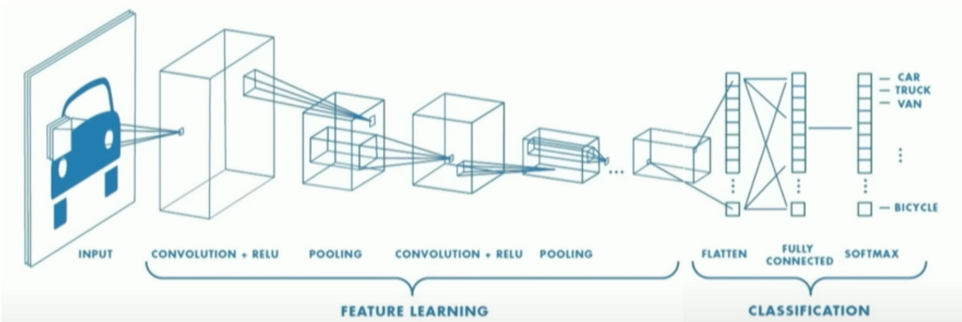
$$\begin{bmatrix} 1 & 2 & 0 & 3 & 1 & 2 \\ 4 & 1 & 0 & 1 & 7 & 3 \\ 1 & 3 & 2 & 2 & 3 & 0 \\ 0 & 1 & 3 & 1 & 1 & 5 \\ 2 & 4 & 1 & 2 & 0 & 1 \\ 1 & 2 & 1 & 3 & 1 & 4 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 7 & 7 \end{bmatrix}$$

When using a stride and no padding in convolution operations, it's standard practice to only compute values for the positions where the filter fits entirely within the input dimensions.

   b. (**5**) Given the same input matrix, compute the resulting feature map using a min pooling operation with a pool size 2x2.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

   c. (**5**) Please draw the CNN feature learning + classification pipeline and give a brief description for the two phases.



Feature Learning:

1. Learn features in input images through convolution
2. Introduce non-linearity through activation function (real-world data is nonlinear)
3. Reduce dimensionality and preserve spatial invariance with pooling

Classification:

1. CONV and POOL layers output high-level features of input
2. Fully connected layers uses those features for classifying input images
3. Express output as probability of an image to belong to a particular class